1.5 Exercises

September 29, 2023

1

The radicand.

2

Consider some fraction with a denominator that is a square root, or is of the form $b+\sqrt{a},\ b-\sqrt{a}$. To fully simplify this fraction, one needs to remove any radicals from that denominator, which is done through a proccess called rationalizing the denominator. In the first case, this can be done by multiplying the fraction by $\frac{\sqrt{b}}{\sqrt{b}}$, and it can be done in the second case by multiplying the fraction by $\frac{a-\sqrt{b}}{a-\sqrt{b}}$, and it can be done in the third case by multiplying the fraction by $\frac{a+\sqrt{b}}{a+\sqrt{b}}$.

3

$\sqrt{28} = \sqrt{7} \times \sqrt{4}$
$= 2\sqrt{7}$

4

$\sqrt{192} = \sqrt{64} \times \sqrt{3}$
$= 8\sqrt{3}$

5

$\sqrt{150} = \sqrt{25} \times \sqrt{6}$
$= 5\sqrt{6}$

6

$\sqrt{3} \times \sqrt{27} = \sqrt{81} = 9$

7

$4\sqrt{6} \times \sqrt{6} = 4 \times 6 = 24$

8

$5\sqrt{24} \times 3\sqrt{10} = 15\sqrt{8}\sqrt{3}\sqrt{5}\sqrt{2} = 15\sqrt{16} \times \sqrt{15} = 60\sqrt{15}$

9

$\sqrt{\frac{5}{16}} = \frac{\sqrt{5}}{\sqrt{16}} = \frac{\sqrt{5}}{4}$

10

$\sqrt{\frac{35}{36}} = \frac{\sqrt{35}}{\sqrt{36}} = \frac{\sqrt{35}}{6}$

11

$\frac{8}{\sqrt{3}} = \frac{8 \times \sqrt{3}}{\sqrt{3}\sqrt{3}} = \frac{8\sqrt{3}}{3}$

12

$$\frac{7}{\sqrt{12}} = \frac{7\sqrt{12}}{\sqrt{12}\sqrt{12}} = \frac{7\sqrt{12}}{12}$$

13

$$\sqrt{\frac{18}{11}} = \frac{\sqrt{18}}{\sqrt{11}} = \frac{3\sqrt{2}\sqrt{11}}{\sqrt{11}\sqrt{11}} = \frac{3\sqrt{22}}{11}$$

14

$$\sqrt{\frac{13}{28}} = \frac{\sqrt{13}}{\sqrt{28}} = \frac{\sqrt{13}}{2\sqrt{7}} = \frac{\sqrt{13}\sqrt{7}}{2\sqrt{7}\sqrt{7}} = \frac{\sqrt{91}}{14}$$

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$$\frac{2}{1-\sqrt{3}} = \frac{2\times(1+\sqrt{3})}{(1-\sqrt{3})\times(1+\sqrt{3})}$$
$$= \frac{2+2\sqrt{3}}{-2} = \frac{1+\sqrt{3}}{-1}$$

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$$\frac{1}{5+\sqrt{6}} = \frac{1\times(5-\sqrt{6})}{(5+\sqrt{6})\times(5-\sqrt{6})} = \frac{5-\sqrt{6}}{25-6} = \frac{5-\sqrt{6}}{19}$$

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$$\frac{\sqrt{2}}{4+\sqrt{5}} = \frac{\sqrt{2}\times(4-\sqrt{5})}{(4+\sqrt{5})\times(4-\sqrt{5})} = \frac{4\sqrt{2}-\sqrt{10}}{11}$$

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$$\frac{3+\sqrt{7}}{2-\sqrt{10}} = \frac{3+\sqrt{7}\times(2+\sqrt{10})}{(2-\sqrt{10})\times(2+\sqrt{10})} = \frac{6+2\sqrt{7}+3\sqrt{10}+\sqrt{70}}{-6}$$

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$\sqrt{108} = \sqrt{36}\sqrt{3} = 6\sqrt{3}$, and thus, the correct answer is C.

20

The answer does not factor out the perfect square factors of 24:
$\sqrt{96} = \sqrt{16}\sqrt{6} = 4\sqrt{6}$

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The error is that the ngeative solution to the quadratic equation was ignored. Instead, the answer is that $x = 9 \vee x = -9$.

22

$s^2 = 169$
$\rightarrow s = \pm 13$

23

$a^2 = 50$
$\rightarrow a = \pm\sqrt{50} = \pm 5\sqrt{2}$

24

$x^2 = 84$
$\rightarrow x = \pm\sqrt{84} = \pm 2\sqrt{21}$

25

$6z^2 = 150$
$\rightarrow z^2 = 25$
$\rightarrow z = \pm 5$

26

$4p^2 = 448$
$\rightarrow p^2 = 112$
$\rightarrow p = \pm\sqrt{112} = \pm 4\sqrt{7}$

27

$-3w^2 = -213$
$\rightarrow w^2 = 71$
$\rightarrow w = \pm\sqrt{71}$

28

$7r^2 - 10 = 25$
$\rightarrow r^2 = 5$
$\rightarrow r = \pm\sqrt{5}$

29

$\frac{x^2}{25} - 6 = -2$
$\rightarrow x^2 = 100$
$\rightarrow x = \pm 10$

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$\frac{t^2}{20} + 8 = 15$
$\rightarrow t^2 = 140$
$\rightarrow t = \pm\sqrt{140}$
$\rightarrow t = \pm 2\sqrt{35}$

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$4(x-1)^2 = 8$
$\rightarrow (x-1)^2 = 2$
$\rightarrow x - 1 = \pm\sqrt{2}$
$\rightarrow x = 1 \pm \sqrt{2}$

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$7(x-4)^2 - 18 = 10$
$\rightarrow (x-4)^2 = 4$
$\rightarrow x - 4 = \pm 2$
$\rightarrow x = 6 \vee x = 2$

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$2(x+2)^2 - 5 = 8$
$\rightarrow (x+2)^2 = \frac{13}{2}$
$\rightarrow x + 2 = \pm\sqrt{\frac{13}{2}}$
$\rightarrow x = \sqrt{\frac{13}{2}} - 2 \vee x = -\sqrt{\frac{13}{2}} - 2$

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(a): $\sqrt{12} - x\sqrt{2} \leq \sqrt{3}$
$\rightarrow x\sqrt{2} \leq \sqrt{3}$
$\rightarrow x \geq \frac{\sqrt{3}}{\sqrt{2}}$

$\rightarrow x \geq \sqrt{\frac{3}{2}}$
(b): $x\sqrt{3} > 2 + x$
$\rightarrow x\sqrt{3} - x > 2$
$\rightarrow (\sqrt{3} - 1)x > 2$
$\rightarrow x > \frac{2}{-1+\sqrt{3}}$
$\rightarrow x > \frac{2(-1-\sqrt{3})}{-2}$
$\rightarrow x > -(-1 - \sqrt{3})$
$\rightarrow x > 1 + \sqrt{3}$
(c): $2\sqrt{5} = 4 + x\sqrt{8}$
$\rightarrow 4 + 2\sqrt{2}x = 2\sqrt{5}$
$\rightarrow 2\sqrt{2}x = 2\sqrt{5} - 4$
$\rightarrow x = \frac{2\sqrt{5}-4}{2\sqrt{2}}$
$\rightarrow= \frac{4\sqrt{10}-8\sqrt{2}}{8}$
$\rightarrow= \frac{\sqrt{10}-2\sqrt{2}}{2}$

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One method is two move 4 to the right side of the equation, and then take its square roots, thus yielding $x = 2 \vee x = -2$. Alternatively, we can factor this equation by using the differences of squares pattern:
$(x + 2)(x - 2) = 0$
$\rightarrow x = 2 \vee x = -2$

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(a): $x^2 = 9$
$\rightarrow x = \pm 3$
(b): $x^2 = 0$
$\rightarrow x = 0$
(c): $x^2 = -4$
$\rightarrow x = \sqrt{-4}$

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$a(x + b)^2 = c$
$\rightarrow (x + b)^2 = \frac{c}{a}$

$\rightarrow x + b = \pm\sqrt{\frac{c}{a}}$
$\rightarrow x + b = \pm\frac{\sqrt{c}}{\sqrt{a}}$
$\rightarrow x + b = \pm\frac{\sqrt{ac}}{a}$
$\rightarrow x = \pm\frac{\sqrt{ac}}{a} - b$

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$h = -16t^2 + 40$
$\rightarrow 0 = -16t^2 + 40$
$\rightarrow t^2 = \frac{5}{2}$
$\rightarrow t = \pm\sqrt{\frac{5}{2}} \approx \pm 1.58$, and thus, the diver is in the air for about 1.58 seconds.

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Earth: $0 = -\frac{32}{2}t^2 + 150$
$\rightarrow t^2 = 9.375$
$\rightarrow t = \pm\sqrt{9.375} \approx \pm 3.062$, meaning it takes about 3.062 seconds for the rock to fall on Earth.
Mars: $0 = -\frac{12}{2}t^2 + 150$
$\rightarrow t^2 = 25$
$\rightarrow t = \pm 5$, meaning it takes 5 seconds for the rock to fall on Mars.
Jupiter: $0 = -\frac{76}{2}t^2 + 150$
$\rightarrow t^2 = \frac{75}{19}$
$\rightarrow t = \pm\sqrt{\frac{75}{19}} \approx \pm 1.97$, and thus, it takes about 1.97 seconds for the rock to fall on Jupiter.
Saturn: $0 = -\frac{30}{2}t^2 + 150$
$\rightarrow t^2 = 10$
$\rightarrow t = \pm\sqrt{10} \approx \pm 3.162$, and thus, it takes about 3.162 seconds for the rock to fall on Saturn.
Pluto: $0 = -\frac{2}{2}t^2 + 150$
$\rightarrow t^2 = 150$
$\rightarrow t = \pm\sqrt{150} \approx \pm 12.247$, and thus, it takes about 12.247 seconds for the rock to fall on Pluto.

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Wind for 5 ft. wave: $5 = 0.019s^2$
$\rightarrow s^2 = \frac{5}{0.019}$
$\rightarrow s = \pm\sqrt{\frac{5}{0.019}} \approx \pm 16.222$
Wind for 20 ft. wave: $20 = 0.019s^2$
$\rightarrow s^2 = \frac{20}{0.019}$
$\rightarrow s = \pm\sqrt{\frac{20}{0.019}} \approx \pm 32.444$
Hence, to produce a wave 20 ft. high, it requires double the wind speed (32.444) as it does to produce a 5 ft. wave (16.222).

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(a): $100 = \pi r^2$
(b): $r^2 = \frac{100}{\pi}$
$\rightarrow r = \pm\sqrt{\frac{100}{\pi}}$
$\rightarrow r = \pm\frac{\sqrt{100}}{\sqrt{\pi}} = \pm\frac{10}{\sqrt{\pi}}$, and thus, because only positive values are accepted, $r = \frac{10}{\sqrt{\pi}} \approx 5.64$ (c): We consider any square with side lengths s. A circle with the same area as this square implies $s^2 = \pi r^2$
$\rightarrow r^2 = \frac{s^2}{\pi}$
$\rightarrow r = \sqrt{\frac{s^2}{\pi}} = \frac{s}{\sqrt{\pi}} = \frac{s\sqrt{\pi}}{\pi}$

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(a): $5 = 0.00829s^2$
$\rightarrow s^2 = \frac{5}{0.00829}$
$\rightarrow s = \pm\sqrt{\frac{5}{0.00829}} \approx \pm 24.56$, and thus, the bicyclist was going about 24.56 mph.
(b): Consider a bicyclist having an air resistance of r and a speed of s. Doubling the speed yields the folliwng equation:
$R = 0.00829(2s)^2 = 4(0.00829s^2)$, and thus, because $r = 0.00829s^2$, doubling the speed quadruples the air resistance.

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To solve this, we set the given equation to 0, and then solve for t:
$0 = (\sqrt{h_0} - \frac{2\pi d^2\sqrt{3}}{lw}t)^2$
$\rightarrow 0 = \sqrt{h_0} - \frac{2\pi d^2\sqrt{3}}{lw}t$
$\rightarrow \frac{2\pi d^2\sqrt{3}}{lw}t = \sqrt{h_0}$
$\rightarrow t = \frac{\sqrt{h_0}}{\frac{2\pi d^2\sqrt{3}}{lw}}$
$\rightarrow = \frac{\sqrt{h_0}lw}{2\pi d^2\sqrt{3}}$
$\rightarrow = \frac{\sqrt{3h_0}lw}{6\pi d^2}$